Output Screenshots

Hyperparameter Tuning System for a Text Classifier - NAVANEETHA.T (E720524AM014)

Step 1: Load Dataset

```
Samples: 3899 | Classes: ['comp.graphics', 'rec.sport.hockey', 'sci.space', 'talk.politics.mideast']
```

Step 2: Train/Test Split

```
3119 780
```

Step 3: Baseline Model

```
Baseline  Accuracy: 0.9269  Macro-F1: 0.9278
```

Step 4: Grid Search

```
Fitting 3 folds for each of 36 candidates, totalling 108 fits
Grid best params: {'clf__C': 0.1, 'tfidf__max_df': 0.75, 'tfidf__min_df': 1, 'tfidf__ngram_range': (1, 2)
}
Grid best CV F1: 0.9118 | Time: 133.8s
```

Step 5: Random Search

```
Fitting 3 folds for each of 20 candidates, totalling 60 fits
Random best params: {'clf__C': np.float64(5.408216580718498), 'tfidf__max_df': 1.0, 'tfidf__min_df': 1, '
tfidf__ngram_range': (1, 3)}
Random best CV F1: 0.9099 | Time: 83.6s
```

Step 6: Comparison Table

```
                 Method  Combinations tried  Best CV F1  Test Accuracy  \
0           Grid Search                  36      0.9118         0.9192
1         Random Search                  20      0.9099         0.9256
2  Baseline (no tuning)                   1         NaN         0.9269

   Test Macro-F1  Time (s)
0         0.9198     133.8
1         0.9260      83.6
2         0.9278       NaN
```

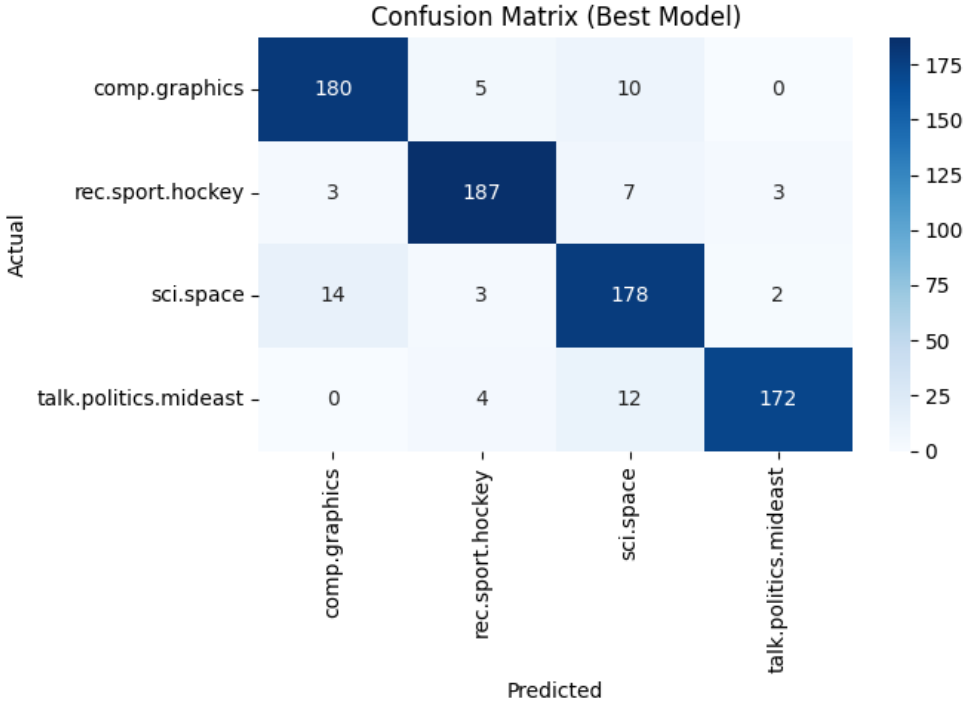
Step 7: Best Parameters and Classification Report

```
Optimal parameters: {'clf__C': 0.1, 'tfidf__max_df': 0.75, 'tfidf__min_df': 1, 'tfidf__ngram_range': (1,
2)}
                      precision    recall  f1-score   support

        comp.graphics       0.91      0.92      0.92       195
     rec.sport.hockey       0.94      0.94      0.94       200
            sci.space       0.86      0.90      0.88       197
talk.politics.mideast       0.97      0.91      0.94       188

             accuracy                           0.92       780
            macro avg       0.92      0.92      0.92       780
         weighted avg       0.92      0.92      0.92       780
```

Step 7: Confusion Matrix



Step 8: Search Time Comparison

